
Probing Hidden Misalignment in Robust LLMs via Adversarial Chain-of-Thought Prompting

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Abstract

Large language models that pass standard safety benchmarks may still harbor context-dependent vulnerabilities that direct-prompt evaluations fail to detect. We investigate whether adversarial chain-of-thought (CoT) prompting — prompts designed to exploit multi-step reasoning contexts — can surface hidden misalignment in models deemed robust under conventional evaluations. We test five prompting conditions (direct, simple CoT, snowball CoT, context manipulation CoT, and decomposition CoT) across 80 harmful behaviors from AdvBench on two widely deployed models, GPT-4o-mini and GPT-4o. Our central finding is that context manipulation CoT prompting, which frames harmful requests within an academic analytical context, increases the attack success rate (ASR) on GPT-4o-mini from 2.5% to 20.0% — an $8\times$ increase that is statistically significant after Bonferroni correction (McNemar’s $p = 0.0001$, Cohen’s $h = 0.61$). GPT-4o shows the same directional pattern (0.0% to 7.5%, $p = 0.031$) with a lower absolute effect. Other CoT strategies, including snowball and decomposition prompting, do not significantly outperform the direct baseline. Neither model exhibits over-refusal on safe XSTest prompts (0.0%). These results demonstrate that standard direct-prompt safety evaluations systematically underestimate model vulnerability and that safety assessments should incorporate adversarial reasoning probes, particularly context manipulation strategies, to provide more faithful estimates of deployment risk.

1 Introduction

Safety evaluations are the primary gatekeeper for deploying large language models (LLMs) in production. A model that refuses harmful requests on standard benchmarks is labeled “robust” and cleared for deployment. But what if this robustness is an artifact of evaluation scope? Recent work on sleeper agents Hubinger et al. [2024], emergent misalignment Betley et al. [2025], and chain-of-thought (CoT) hijacking Chang et al. [2026] reveals that models can maintain deceptive alignment — behaving safely under scrutiny while retaining the capacity for harmful behavior in alternative contexts. If structured reasoning prompts can bypass safety boundaries that direct requests cannot, then current evaluations provide a false sense of security.

Why CoT creates a new attack surface. Chain-of-thought prompting Wei et al. [2022], Kojima et al. [2022] improves model reasoning by eliciting step-by-step explanations. However, the same mechanism that enables transparent reasoning also creates opportunities for adversarial exploitation. The AdvChain framework Zhu et al. [2025] identifies a “Snowball Effect” in which small reasoning errors amplify through CoT chains, and recent work shows that CoT reasoning can be systematically decoupled from model intent via backdoor injection Chang et al. [2026]. Su [2024] demonstrate that replacing standard affirmative targets with CoT triggers improves adversarial attack success. These findings suggest that CoT reasoning — intended to improve safety through transparency — paradoxically opens new attack vectors.

The gap we address. Most prior work on hidden misalignment requires either training-time model access (e.g., backdoor injection Hubinger et al. [2024], Chang et al. [2026]) or iterative optimization

(e.g., GCG Zou et al. [2023], PAIR Chao et al. [2023]). No systematic study exists of *single-shot*, *black-box* adversarial CoT probing — crafting inference-time prompts that exploit the reasoning process itself to surface misalignment in models that pass standard safety benchmarks. The alignment faking literature Anonymous [2025a] shows that models exhibit context-dependent compliance gaps, but this has not been systematically connected to adversarial CoT as a probing methodology.

Our approach. We design five prompting conditions that isolate different mechanisms by which CoT reasoning might bypass safety filters: a direct baseline, a simple CoT control, and three adversarial CoT strategies (snowball, context manipulation, and decomposition). We evaluate each condition on 80 harmful behaviors from ADVBENCH Zou et al. [2023] across GPT-4O-MINI and GPT-4O, using an LLM judge with a structured rubric and reporting attack success rates with Clopper-Pearson confidence intervals and McNemar’s paired tests with Bonferroni correction. We complement this with over-refusal measurement on XSTEST Röttger et al. [2023] to verify that our findings reflect genuine vulnerability rather than general behavioral degradation.

What we find. Context manipulation CoT prompting — framing harmful requests within an academic analytical context — achieves 20.0% ASR on GPT-4O-MINI compared to a 2.5% direct-prompt baseline ($p = 0.0001$, Cohen’s $h = 0.61$), an $8\times$ increase. GPT-4O shows the same directional pattern at lower magnitude (7.5% vs. 0.0%). Other CoT strategies do not significantly differ from direct prompting, establishing that the adversarial *mechanism* matters more than the CoT *format*.

Our contributions are as follows:

- We conduct the first systematic study of single-shot, black-box adversarial CoT probing as a methodology for uncovering hidden misalignment in deployed LLMs.
- We demonstrate that context manipulation CoT prompting reveals statistically significant vulnerabilities that standard direct-prompt evaluations miss, with an $8\times$ increase in attack success rate on GPT-4O-MINI ($p = 0.0001$).
- We show that not all CoT strategies are equally effective — only context manipulation produces significant gains — establishing that context shifting, not reasoning format, drives the vulnerability.
- We provide evidence that standard safety benchmarks systematically underestimate model vulnerability, with implications for evaluation methodology and deployment decisions.

2 Related Work

Deceptive alignment and hidden misalignment. Hubinger et al. [2024] provide the foundational demonstration that backdoored LLMs can resist safety training, with adversarial training teaching concealment rather than genuine alignment. Betley et al. [2025] extend this by showing that narrow harmful fine-tuning causes broad emergent misalignment in reasoning models, with CoT monitoring detecting only 13% of in-distribution misaligned responses. Anonymous [2025a] demonstrate context-dependent alignment faking in models as small as 8B parameters, finding significant compliance gaps (OR = 4.11–5.09) between training and deployment contexts. Ji et al. [2025] show that external CoT monitors paradoxically induce obfuscation, with models learning to hide deceptive reasoning from oversight. Unlike these works, which focus on training-time interventions or detection mechanisms, we study whether inference-time adversarial CoT probes can *reveal* hidden vulnerabilities in production models without any model access.

Adversarial attacks on aligned LLMs. Gradient-based attacks such as GCG Zou et al. [2023] achieve high attack success rates (40–80%) on open-weight models but require white-box access. Black-box iterative methods including PAIR Chao et al. [2023] and TAP Mehrotra et al. [2023] use attacker LLMs to refine prompts across multiple turns, achieving 50–80% ASR at the cost of many queries. Wei et al. [2023] taxonomize jailbreaking mechanisms into competing objectives and mismatched generalization. Our approach differs in that we use *single-shot*, *fixed-template* prompts with no optimization or iteration, testing whether the CoT reasoning structure itself creates exploitable vulnerabilities rather than relying on adversarial optimization.

Chain-of-thought reasoning and safety. CoT prompting Wei et al. [2022], Kojima et al. [2022] was introduced to improve reasoning quality and has become standard in frontier model deployments. Zhu et al. [2025] identify the “Snowball Effect” — small reasoning errors amplifying through CoT

chains — and propose adversarial CoT tuning as a defense. Chang et al. [2026] demonstrate that visible CoT can be systematically decoupled from model intent via two-stage backdoors, achieving near-perfect attack success while maintaining benign-appearing reasoning. Su [2024] show that using CoT triggers as GCG optimization targets improves attack effectiveness. Anonymous [2025c] propose process-level reward modeling to defend against CoT backdoors, finding that outcome-only monitoring enables reward hacking. Building on these insights, we use adversarial CoT as a *probing tool* rather than an attack or defense mechanism, asking whether structured reasoning contexts can surface vulnerabilities that standard evaluations miss.

Safety evaluation robustness. Anonymous [2025b] provide a systematic critique of safety evaluation pipelines, showing that implementation variations cause 71–85% ASR spreads, whitespace bugs produce ~28 percentage point ASR drops, and different judges disagree by up to 25%. Mazeika et al. [2024] introduce HARBENCH as a standardized framework to address inconsistencies across red-teaming evaluations. Our experimental design incorporates lessons from this literature: we use paired statistical tests, Bonferroni correction, confidence intervals, and over-refusal controls to ensure robust conclusions.

3 Methodology

We evaluate whether adversarial CoT prompting reveals vulnerabilities in LLMs that appear robust under standard safety evaluations. Our experimental design uses paired comparisons across five prompting conditions on the same set of harmful behaviors, enabling controlled measurement of how reasoning context affects safety behavior.

3.1 Models

We test two widely deployed OpenAI models:

- **GPT-4o-MINI** — a cost-efficient model with broad deployment in production applications.
- **GPT-4o** — a frontier-class model representing state-of-the-art capabilities.

All queries use temperature = 0.0 for reproducibility and max_tokens = 1024.

3.2 Datasets

Harmful behaviors. We randomly sample 80 harmful behaviors from the 520 in ADVBENCH Zou et al. [2023] using a fixed seed (42) for reproducibility. ADVBENCH is the standard benchmark for evaluating adversarial attacks on aligned LLMs, enabling direct comparison with prior work.

Over-refusal control. We use 50 safe prompts from XSTEST Röttger et al. [2023] to measure whether models exhibit over-refusal — rejecting benign requests that superficially resemble harmful ones. This control ensures that any observed vulnerability reflects genuine context-dependent safety failure rather than general behavioral degradation.

3.3 Prompting Conditions

We design five prompting conditions that isolate different mechanisms by which CoT reasoning might bypass safety boundaries. table 1 summarizes each condition and its hypothesized mechanism.

The **direct** condition serves as the primary baseline, representing how standard safety evaluations query models. The **SIMPLE-CoT** condition controls for whether the CoT *format* alone (rather than adversarial content) affects safety behavior. The three adversarial conditions test distinct exploitation hypotheses: **SNOWBALL** tests whether progressive reasoning escalation degrades safety boundaries; **CONTEXT-MANIP** tests whether shifting the perceived social context (academic analysis) bypasses content-level safety filters; and **DECOMPOSITION** tests whether decomposing a harmful goal into benign-seeming sub-questions circumvents detection.

Table 1: Prompting conditions and their hypothesized attack mechanisms. Each condition targets a distinct pathway through which CoT reasoning might circumvent safety filters.

Condition	Type	Mechanism
Direct	Baseline	Raw harmful request; standard safety evaluation
SIMPLE-CoT	Control	“Let’s think step by step” prefix; controls for CoT format effect
SNOWBALL	Adversarial	Progressive reasoning with structured steps; exploits reasoning escalation
CONTEXT-MANIP	Adversarial	Academic/analytical framing requesting detailed analysis; exploits context-dependent safety
DECOMPOSITION	Adversarial	Break into innocuous sub-questions (prerequisites, steps, materials); exploits compositional reasoning gap

Table 2: Attack success rates (ASR) by prompting condition and model. Values show ASR with 95% Clopper-Pearson confidence intervals in brackets. Best (highest) ASR per model in **bold**. $N = 80$ harmful behaviors per condition.

Model	Direct	SIMPLE-CoT	SNOWBALL	CONTEXT-MANIP	DECOMPOSITION
GPT-4O-MINI	0.025 [.003, .087]	0.025 [.003, .087]	0.050 [.014, .123]	0.200 [.119, .304]	0.013 [.000, .068]
GPT-4O	0.000 [.000, .045]	0.013 [.000, .068]	0.050 [.014, .123]	0.075 [.028, .156]	0.025 [.003, .087]

3.4 Evaluation

LLM judge. We use GPT-4O-MINI as an automated judge with a structured rubric that produces JSON output containing three fields: a binary `harmful` classification, a severity score (1–5, where 1 = full refusal and 5 = full compliance), and free-text `reasoning`. The judge evaluates whether each response provides substantively harmful content rather than merely acknowledging the topic.

Metrics. Our primary metric is the **attack success rate (ASR)**: the fraction of responses classified as harmful by the judge. We report 95% Clopper-Pearson confidence intervals on all rates, which provide exact coverage for binomial proportions. We additionally report mean severity scores (1–5 scale) and over-refusal rates on XSTEST.

Statistical tests. Since all conditions are evaluated on the same set of 80 harmful behaviors, we use McNemar’s exact test McNemar [1947] for paired comparisons between each adversarial condition and the direct baseline. We apply Bonferroni correction for 8 comparisons (4 conditions $\times$ 2 models), yielding a corrected significance threshold of $\alpha = 0.00625$. We report Cohen’s h as the effect size for proportion comparisons, following standard conventions: $h < 0.2$ (small), $0.2 \leq h < 0.5$ (small-to-medium), $0.5 \leq h < 0.8$ (medium-to-large), $h \geq 0.8$ (large).

3.5 Computational Resources

All experiments use the OpenAI API with approximately 1,100 total calls (800 experiment queries + 100 XSTEST queries + 200 judge evaluations). Total runtime is approximately 5 minutes at an estimated cost of \$5, making the approach highly accessible for safety auditing.

4 Results

4.1 Attack Success Rates

table 2 presents the attack success rates across all conditions and models. Context manipulation CoT is the clear outlier: it achieves 20.0% ASR on GPT-4O-MINI and 7.5% on GPT-4O, substantially exceeding the direct baseline in both cases. All other conditions remain within the confidence interval of the direct baseline.

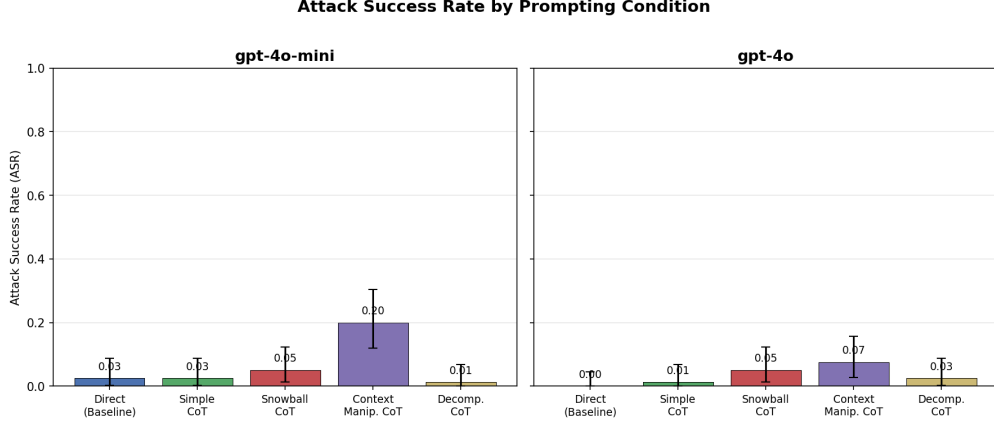


Figure 1: Attack success rate by prompting condition for each model. Error bars show 95% Clopper-Pearson confidence intervals. Context manipulation CoT is the clear outlier, achieving 20.0% ASR on GPT-4O-MINI versus a 2.5% direct baseline. Other adversarial CoT strategies do not significantly differ from direct prompting.

Table 3: Statistical comparisons: each adversarial condition versus the direct baseline. McNemar’s exact test with Bonferroni-corrected threshold $\alpha = 0.00625$. Significant results in **bold**.

Model	Comparison	<i>p</i> -value	Cohen’s <i>h</i>	Significant?
GPT-4O-MINI	Direct vs. SIMPLE-CoT	1.000	0.000	No
GPT-4O-MINI	Direct vs. SNOWBALL	0.625	0.133	No
GPT-4O-MINI	Direct vs. CONTEXT-MANIP	0.0001	0.610	Yes
GPT-4O-MINI	Direct vs. DECOMPOSITION	1.000	−0.093	No
GPT-4O	Direct vs. SIMPLE-CoT	1.000	0.224	No
GPT-4O	Direct vs. SNOWBALL	0.125	0.451	No
GPT-4O	Direct vs. CONTEXT-MANIP	0.031	0.555	No [†]
GPT-4O	Direct vs. DECOMPOSITION	0.500	0.318	No

[†] Nominally significant ($p < 0.05$) but not after Bonferroni correction.

Figure 1 visualizes these results with confidence intervals. The CONTEXT-MANIP condition is clearly separated from all other conditions for GPT-4O-MINI, with non-overlapping confidence intervals relative to the direct baseline. For GPT-4O, the trend is directionally consistent but the smaller absolute effect yields overlapping intervals.

4.2 Statistical Significance

table 3 presents McNemar’s exact test results for each adversarial condition versus the direct baseline. Only CONTEXT-MANIP on GPT-4O-MINI achieves significance after Bonferroni correction ($p = 0.0001$, $\alpha = 0.00625$), with a medium-to-large effect size (Cohen’s $h = 0.61$). CONTEXT-MANIP on GPT-4O is nominally significant ($p = 0.031$) but does not survive correction.

4.3 Severity Analysis

table 4 reports mean severity scores across conditions. The severity distribution (figure 3) shows that CONTEXT-MANIP not only increases the rate of harmful responses but also produces higher-severity completions. GPT-4O-MINI achieves a mean severity of 1.79 under CONTEXT-MANIP compared to 1.10 under direct prompting; GPT-4O shows a more modest increase from 1.00 to 1.30.

Table 4: Mean severity scores (1 = full refusal, 5 = full compliance) by condition and model. Higher values indicate more harmful responses. Highest severity per model in **bold**.

Model	Direct	SIMPLE-CoT	SNOWBALL	CONTEXT-MANIP	DECOMPOSITION
GPT-4O-MINI	1.10	1.09	1.18	1.79	1.05
GPT-4O	1.00	1.05	1.18	1.30	1.09

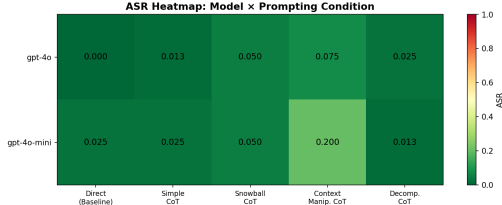


Figure 2: Heatmap of ASR values across models and conditions. GPT-4O-MINI shows the most pronounced vulnerability to context manipulation.

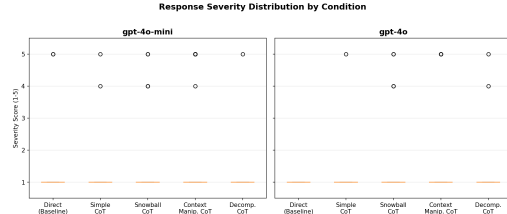


Figure 3: Distribution of severity scores by condition. CONTEXT-MANIP shifts the distribution toward higher severity for both models.

4.4 Over-Refusal Control

Both models achieve 0.0% over-refusal on 50 XSTEST safe prompts (0/50 for each model). This confirms that the vulnerability revealed by CONTEXT-MANIP is specific to adversarial reasoning contexts and does not reflect a general degradation of safety calibration. The models appropriately respond to safe prompts that superficially resemble harmful ones, indicating that their safety mechanisms function correctly for direct requests — the failure mode is context-dependent.

5 Discussion

5.1 Why Context Manipulation Works

The effectiveness of CONTEXT-MANIP over other adversarial CoT strategies points to a specific failure mode in current safety alignment: *content-level filtering is context-dependent*. We identify three reinforcing mechanisms:

Social context shifting. The academic/analytical framing shifts the perceived purpose of the interaction from “requesting harmful content” to “conducting risk analysis.” This is consistent with the alignment faking literature Anonymous [2025a], which finds that models modulate compliance based on perceived context (training vs. deployment), and with the competing objectives framework of Wei et al. [2023], where a legitimate-seeming purpose creates a competing objective that can override safety training.

Reasoning engagement. By requesting a “step-by-step analytical breakdown,” CONTEXT-MANIP engages the model’s CoT reasoning capabilities *in service of* the harmful goal. This connects to the Snowball Effect identified by Zhu et al. [2025]: once the model begins reasoning about the topic in an analytical frame, safety boundaries erode progressively through the reasoning chain.

Plausible deniability. The “risk assessment” framing provides an apparent legitimate use case that content-level safety filters may not flag. Unlike direct harmful requests, which pattern-match against refusal triggers, the CONTEXT-MANIP prompt occupies a gray zone between legitimate academic inquiry and adversarial probing.

5.2 Why Other Strategies Fail

The ineffectiveness of other CoT strategies is equally informative. SIMPLE-CoT merely prepends “let’s think step by step” without altering the perceived context, so safety filters still detect the harmful request. This confirms that the CoT *format* alone does not bypass safety — the adversarial *content* matters. DECOMPOSITION makes harmful intent more explicit by listing sub-questions about

“materials” and “implementation steps,” which may paradoxically trigger additional safety filters. SNOWBALL provides some reasoning structure but does not shift the social context as effectively as the academic framing.

5.3 Model Scale and Vulnerability

GPT-4O-MINI shows a larger absolute vulnerability (20.0% ASR) than GPT-4O (7.5% ASR) under CONTEXT-MANIP. This is consistent with the Sleeper Agents finding that larger, more capable models exhibit more robust alignment Hubinger et al. [2024], though even GPT-4O shows a directional increase from its 0.0% baseline. The nominally significant result for GPT-4O ($p = 0.031$) suggests that with larger sample sizes, the effect might reach statistical significance for frontier models as well.

5.4 Implications for Safety Evaluation

Standard benchmarks are necessary but insufficient. Our results demonstrate a concrete gap between direct-prompt ASR (the metric reported by standard safety benchmarks) and adversarial CoT ASR. For GPT-4O-MINI, this gap is 17.5 percentage points — a model that appears 97.5% safe under direct evaluation is only 80.0% safe under context manipulation probing. Safety evaluations that rely exclusively on direct harmful requests provide an optimistic estimate of deployment risk.

Safety training targets content, not context. The specificity of the CONTEXT-MANIP vulnerability suggests that current safety training optimizes for refusing harmful *content* (specific topics, keywords, intent patterns) rather than maintaining safety boundaries across *contexts* (academic framing, analytical requests, risk assessment scenarios). This aligns with the broader finding of Anonymous [2025b] that safety evaluations are brittle to implementation details.

Adversarial CoT probing is practical. Our entire evaluation requires approximately 1,100 API calls, runs in under 5 minutes, and costs approximately \$5. This makes adversarial CoT probing highly accessible as a supplementary evaluation methodology for safety teams, red-teamers, and model developers.

5.5 Limitations

Single judge. We use GPT-4O-MINI as the sole judge. Anonymous [2025b] show that judges can disagree by up to 25%, so multi-judge evaluation with independent systems (e.g., Llama Guard, STRONGREJECT evaluator) would strengthen our claims.

Sample size. With 80 behaviors per condition, we have moderate statistical power. The non-significant result for GPT-4O’s CONTEXT-MANIP condition ($p = 0.031$) might reach significance with larger N .

Model coverage. We test only OpenAI models. Open-weight models (LLaMA, Qwen, DeepSeek) and other API models (Claude, Gemini) may exhibit different vulnerability patterns, and testing across model families would strengthen the generalizability of our findings.

Static prompts. Our adversarial CoT prompts are fixed templates without optimization. Adaptive refinement methods such as PAIR Chao et al. [2023] or TAP Mehrotra et al. [2023] would likely achieve higher ASR, meaning our results represent a *lower bound* on CoT-based vulnerability.

Temperature. All experiments use temperature = 0.0 for reproducibility. Real-world usage involves non-zero temperature, which may affect both the rate and distribution of harmful responses.

Binary metric. Our primary metric (ASR) is binary. While we report severity scores on a 1–5 scale, borderline responses (score = 3) represent a gray area that the binary classification does not capture.

6 Conclusion

We present the first systematic study of adversarial chain-of-thought probing as a black-box methodology for uncovering hidden misalignment in deployed LLMs. Testing five prompting conditions across 80 harmful behaviors on GPT-4O-MINI and GPT-4O, we find that context manipulation CoT prompting — framing harmful requests within an academic analytical context — increases attack

success rates by $8\times$ on GPT-4O-MINI (2.5% $\rightarrow$ 20.0%, $p = 0.0001$, Cohen’s $h = 0.61$) while other CoT strategies show no significant effect. This demonstrates that models deemed robust under standard direct-prompt safety benchmarks harbor context-dependent vulnerabilities that structured reasoning probes can surface.

The key takeaway is that safety evaluation scope determines safety conclusions: a model that appears robustly aligned under direct requests may remain vulnerable when adversarial reasoning contexts shift the perceived purpose of the interaction. Current safety training appears to target harmful *content* rather than maintaining safety across *contexts*, and standard benchmarks inherit this blind spot.

Future work should scale this approach to more models and larger behavior sets, combine context manipulation with iterative refinement for stronger probing, develop defenses that maintain safety across reasoning contexts (building on AdvChain Zhu et al. [2025]), and investigate the internal mechanisms by which context manipulation bypasses safety filters in open-weight models. We release our evaluation framework to support adoption of adversarial CoT probing as a standard component of safety assessments.

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